Distributed Threshold Inter-Mesh Theory (D-TIT): A Layered, Conducting-Threshold Architecture for Consciousness

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Hypothesis Paper — Frontier Architecture for Consciousness Science

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Abstract

We propose Distributed Threshold Inter-Mesh Theory (D-TIT), a novel architectural hypothesis of consciousness. In contrast to Global Workspace Theory (GWT) and its neuronal implementations, which posit competition among parallel specialized processors for entry into a limited-capacity broadcast stage, D-TIT models the substrate as a base network embedded in a plane whose nodes possess continuous height values representing attention coefficients. A conducting threshold plane at height *b* induces automatic meshing among all supra-threshold peaks, analogous to electrical contact. Consciousness is layered: multiple parallel planes and intermediate thresholds support functionally differentiated yet non-segregated sub-systems (e.g., threat response linked to amygdala circuitry). The overall phenomenological state is formalized as a stacked vector of layer activations. Biophysical grounding is provided by treating neurons as bio-batteries whose electromagnetic fields and emergent brainwaves implement conductivity and meshing. The theory remains at the hypothesis stage; a complete account is expected to require still-unresolved quantum processes in neural tissue. We supply a mathematical skeleton, biological correspondences, distinguishing predictions, and open problems.

1. Introduction

Consciousness remains incompletely explained by existing frameworks. Global Workspace Theory (Baars, 1988; Baars, 1997) and the Global Neuronal Workspace (Dehaene et al., 1998; Dehaene & Naccache, 2001; Dehaene & Changeux, 2011) successfully account for limited capacity, reportability, and the contrast between local unconscious processing and global availability. Nevertheless, the postulation of a relatively privileged workspace or ignition stage has been criticized as overly centralized relative to the brain's distributed, recurrent architecture.

D-TIT begins from a different geometric and dynamical premise: consciousness arises from thresholded, inter-meshed layers within a continuous height field defined over a base network, rather than from selection into a single broadcast medium. The present paper formalizes this hypothesis, supplies a minimal mathematical structure, maps it onto known biophysics, and identifies the quantum-neural frontier as the probable locus of a complete solution.

2. Critique of the Parallel-Processor-plus-Stage Architecture

GWT and GNW correctly emphasize massive parallelism of specialized processors and the necessity of selection. However, the further claim that selected content must enter a distinct workspace in order to become conscious is not obligatory. An alternative is that selection itself, realized as the crossing of one or more dynamical thresholds, simultaneously creates both local illumination and a new inter-meshed layer among the selected elements. No separate stage is required; the mesh that forms above threshold *is* the conscious layer.

3. Formal Framework of D-TIT

3.1 Base Network and Height Field

Let the cortical (and relevant subcortical) substrate be represented by a weighted undirected graph $G = (V, E, W)$, where V is the set of nodes (neurons, columns, or parcels), E the anatomical or functional edges, and $W: E \to$ ■■ edge weights.

Assign to each node a time-dependent height (attention coefficient)

$$h: V \times \blacksquare\blacksquare \to \blacksquare\blacksquare , \quad h(v, t) \geq 0.$$

The function $h(\cdot, t)$ defines a continuous landscape over the discrete network.

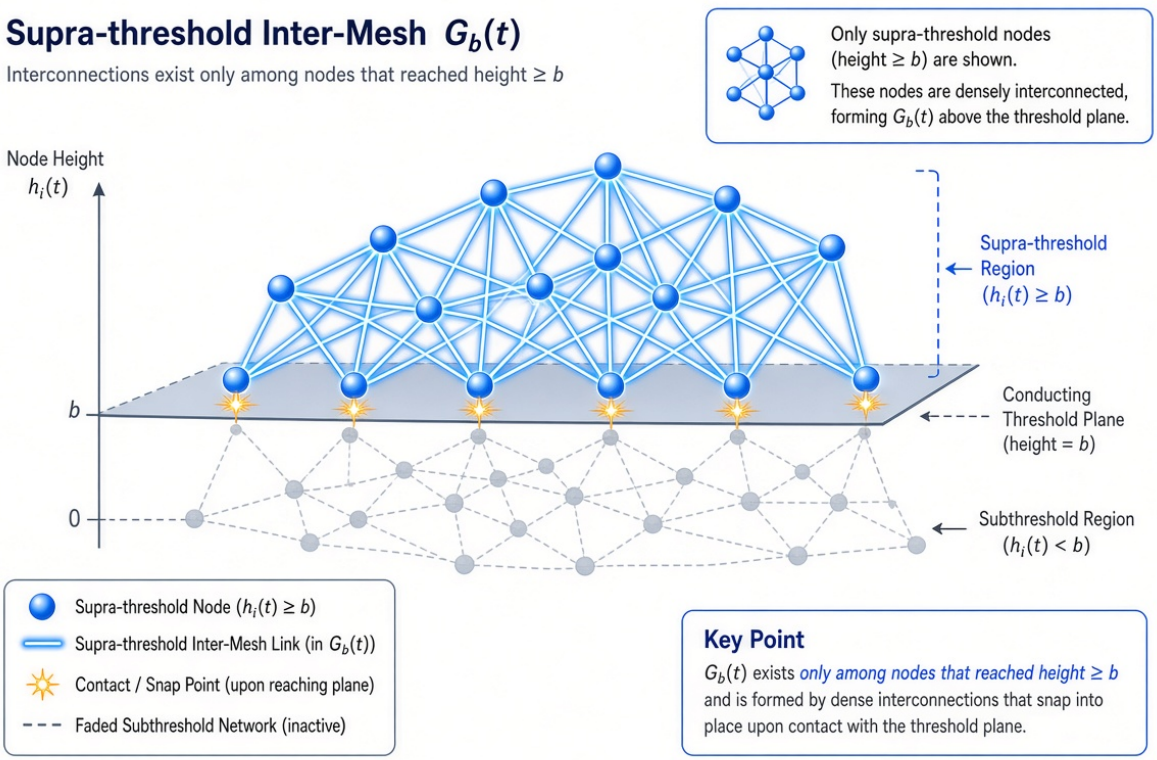


Figure 1. *Base Network and Height Landscape. The underlying anatomical/functional network lies in a 2D plane. Each node carries a continuous height value h(v) representing its current attention coefficient, producing a landscape of peaks and troughs.*

3.2 Conducting Threshold Plane

Fix a threshold value $b > 0$. Define the indicator

$$\sigma_b(v, t) = 1 \quad \text{if } h(v, t) \geq b, \quad \text{otherwise } 0.$$

The supra-threshold set at time t is $V_b(t) = \{ v \in V \mid \sigma_b(v, t) = 1 \}$.

Meshing rule (conducting-plane contact). When a node reaches height b, it is treated as making electrical contact with a conducting plane. All nodes in $V_b(t)$ therefore become mutually coupled with effective conductance determined by their pairwise anatomical weights and instantaneous electromagnetic interaction strength. The induced conscious layer is the subgraph

$$G_b(t) = (V_b(t), E_b(t)), \quad E_b(t) = \{ (u, v) \in E \mid u, v \in V_b(t) \}.$$

Additional edges may be transiently added via field-mediated (ephaptic) coupling.

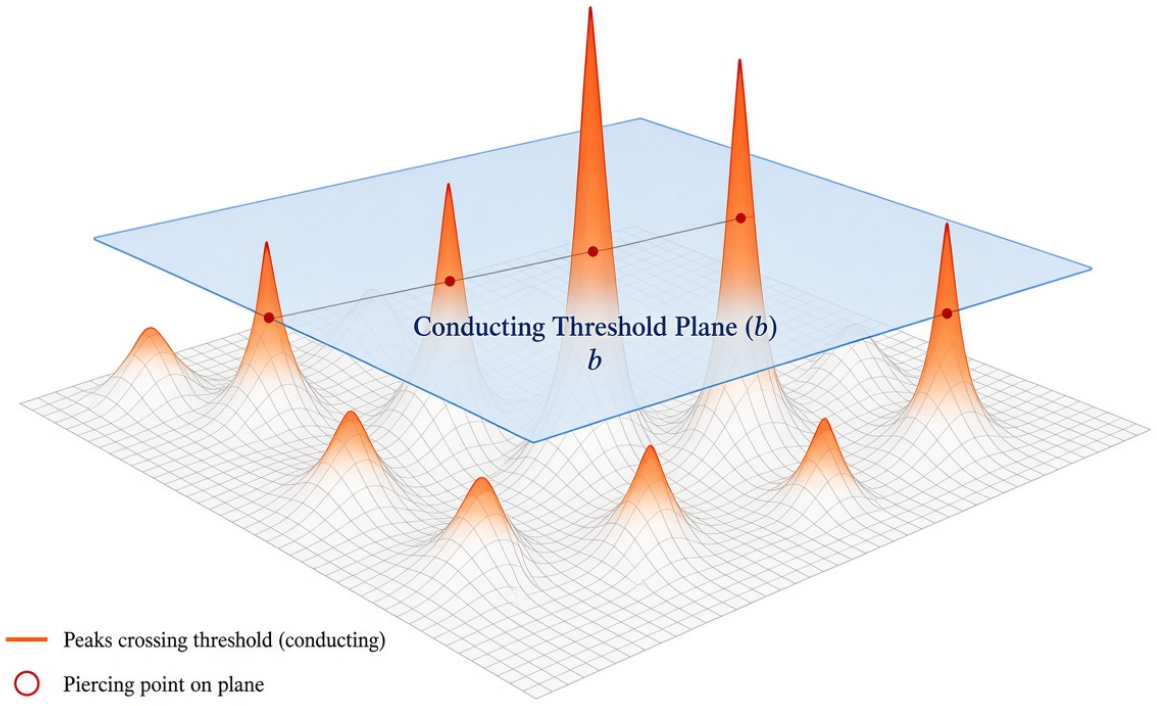


Figure 2. *Conducting Threshold Plane. A horizontal conducting plane sits at height b. Peaks whose attention coefficient reaches or exceeds b pierce the plane (marked). Contact with the plane is the selection event that enables meshing.*

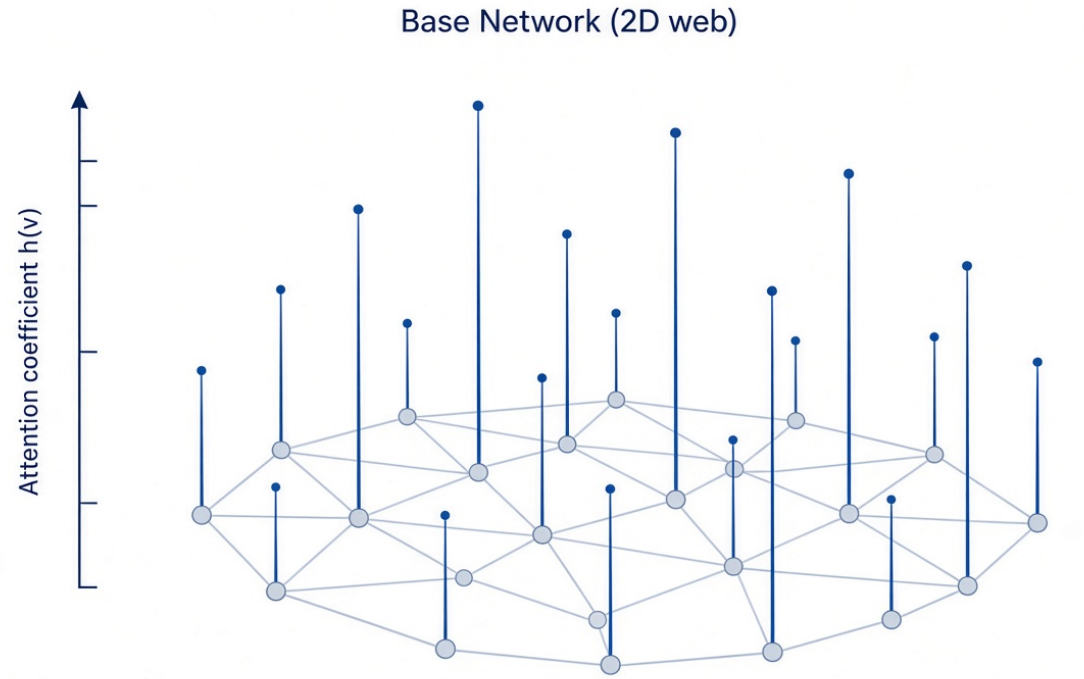


Figure 3. *Formation of the Supra-threshold Inter-Mesh $G_b(t)$. Only nodes that have reached height ≥ b participate. Upon contact with the conducting plane these nodes automatically form dense interconnections (the inter-mesh). Subthreshold nodes remain in the faded base network and do not belong to the conscious layer.*

3.3 Multi-layer Extension

Introduce a finite stack of thresholds $0 < b_1 < b_2 < \ldots < b_K$. Each interval defines a distinct functional plane or sub-system. Planes remain linked by graded connection strengths to specific anatomical targets (e.g., high-threat

peaks preferentially drive amygdala-centered circuits). Segregation is never complete; cross-talk is obligatory.

3.4 Phenomenological State Vector

The instantaneous conscious state is the stacked vector

$$\Psi(t) = (\mathbf{a}_1(t), \mathbf{a}_2(t), \ldots, \mathbf{a}_K(t)),$$

where $\mathbf{a}_k(t)$ is the activation pattern (or a low-dimensional summary) of the mesh belonging to layer k. The qualitative character of experience is identified with the geometry and dynamics of $\Psi(t)$. Global integrative effects analogous to a workspace can emerge at the level of the full vector without requiring a privileged anatomical stage.

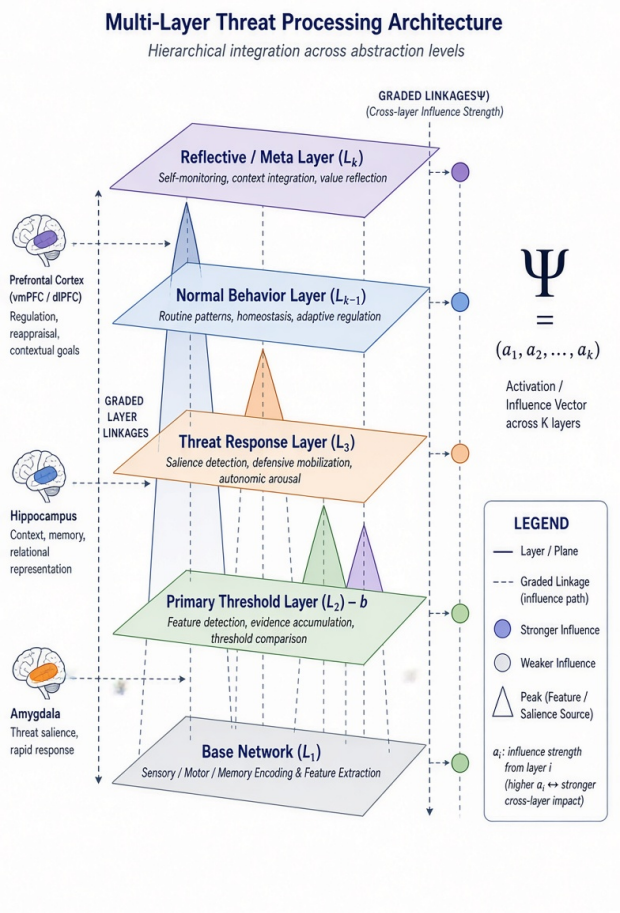


Figure 4. *Multi-layer Stacked Architecture. Multiple parallel planes support functionally differentiated yet interconnected layers (threat response, normal behaviour, reflective/meta processes, etc.). Graded linkages connect layers to specific brain systems. The overall phenomenological state is the stacked vector* $\Psi = (a_1, a_2, \ldots, a_K)$.

4. Biophysical Grounding

Neurons maintain electrochemical gradients and function as microscopic bio-batteries. Transmembrane currents generate local field potentials; synchronous populations produce measurable electromagnetic fields and oscillatory brain rhythms. In D-TIT these fields implement the conductivity of the threshold plane and the automatic meshing of supra-threshold peaks. Ephaptic interactions and oscillatory coherence supply the physical substrate for rapid, selective coupling once $h(v) \geq b$.

Fixed or slowly adapting regional thresholds correlate with known functional specializations (amygdala for rapid defensive mobilization, language networks for report, prefrontal circuits for flexible control). Crossing a region-linked threshold routes content toward the corresponding effector systems while the inter-mesh preserves integration.

5. Relation to Existing Theories

D-TIT shares with GWT/GNW the recognition of selection and limited capacity, yet relocates the integrative step from a workspace into the threshold-induced mesh itself. It is compatible with dynamical-systems accounts of

ignition and with field-based theories that emphasize electromagnetic contributions. It differs from Integrated Information Theory by prioritizing explicit threshold geometry and layered meshes over a scalar complexity measure.

6. Open Problems and the Quantum Frontier

The present formulation is deliberately incomplete. Critical open questions include:

• the precise biophysical variable that corresponds to height $h(v)$ (membrane potential, synaptic drive, local-field amplitude, phase synchrony, …);
• the quantitative rule that converts contact with the conducting plane into meshing conductance;
• the dynamical equations governing the evolution of $h(v, t)$;
• a full account of reportability and flexible control from the stacked vector Ψ.

We conjecture that a satisfactory closure of these gaps will require quantum processes operating in neural tissue (possibly within microtubules, at membrane proteins, or in collective electromagnetic modes). Until the quantum-neural problem is solved, D-TIT remains a classical architectural hypothesis whose ultimate physical implementation is still unknown.

7. Predictions

1. Conscious contents should correspond to connected components of the supra-threshold mesh $G_b(t)$ rather than to a single fronto-parietal ignition pattern.
2. Multi-focal awareness should appear as multiple simultaneous connected components above b, with measurable cross-component field coupling.
3. Graded or intermediate conscious states should map onto partial occupation of intermediate threshold bands.
4. Perturbations that selectively disrupt field-mediated (ephaptic) coupling while sparing spike transmission should impair meshing and therefore conscious access more than unconscious processing.

8. Conclusion

Distributed Threshold Inter-Mesh Theory offers a geometrically explicit, multi-layered alternative to stage-based accounts of consciousness. By treating threshold contact as both selection and meshing mechanism, and by representing phenomenology as a stacked vector of inter-meshed layers, it preserves the functional insights of Global Workspace Theory while embedding them in a fully distributed substrate. The hypothesis is offered as a scaffold for further mathematical, biophysical, and ultimately quantum-neural development.

References

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Additional literature on ephaptic coupling, cortical electromagnetic fields, and quantum models of consciousness (Orch-OR and related proposals) provides further context for the biophysical and quantum sections of the present hypothesis.